Tertiary Infotech Academy Pte Ltd

# Lab 12 — Join, Group and Pivot a Loan Portfolio

Python Programming for Finance · TGS-2026065049 · LO6

## Scenario

A risk team needs exposure and approval metrics by sector, region and risk band from separate application and reference tables.

## Goal

A validated merge, groupby summary and pivot table exported to CSV.

## Tools

Pandas, CSV

## Step-by-step

**1.** Profile application and sector reference keys.

**2.** Merge with validate='many_to_one' and indicator=True.

**3.** Review unmatched rows before dropping or fixing them.

```
python starter.py
```

**4.** Group by sector and calculate count, exposure and mean credit score.

**5.** Create a region-by-risk-band exposure pivot.

```
python solution.py
```

**6.** Reconcile grouped and pivot totals to the merged portfolio.

```
python -m unittest test_solution.py
```

## Acceptance criteria

No unexpected many-to-many join occurs, unmatched keys are reported, and summary exposure totals equal the source.

## Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.